

3D reconstruction of arbitrary granular media utilizing vision foundation model

Ruidong LI , Zhen-Yu YIN ^{*}, Shao-Heng HE

Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Hong Kong, China

HIGHLIGHTS

- VFM is incorporated for the first time in 3D reconstruction of arbitrary granular media.
- The effectiveness of the method is validated on two carbonate sand and four lentil sample datasets.
- Reconstruction accuracy is improved by 50 % compared to the watershed segmentation.
- Over- and under-segmentation errors are minimized significantly.

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ABSTRACT

Reconstructing three-dimensional (3D) granular microstructures through X-ray micro-computed tomography (μ CT) imaging is significant for elucidating micromechanical behaviors of granular media and optimizing geotechnical designs. However, due to the irregular morphology and dense packing of granular media, traditional image-processing techniques often lack the precision required for accurate reconstructions. This paper presents a novel framework for accurate 3D reconstruction of arbitrary granular media using vision foundation models (VFMs). Two-dimensional (2D) mask maps representing the granular media are extracted from μ CT images along the x , y , and z -axes using VFMs and then processed by a two-step strategy to repair textures and remove noises. An optimal transport (OT)-based method is employed to reconstruct a complete 3D mask map based on 2D mask maps. The proposed method is applied to reconstruct two carbonate sand samples with irregular grain shapes and four lentil samples composed of nearly 10,000 grains captured in triaxial loading, utilizing various VFMs and prompt configurations. The framework demonstrated a 50 % improvement in reconstruction accuracy over the state-of-the-art method for carbonate sands and achieved a 95 % accuracy for lentil samples. This advancement offers a more effective alternative for investigating the micromechanics of granular media.

1. Introduction

Granular media are widespread in nature, encompassing a myriad of materials such as sugar, sand, gravel, and even cosmetics [1]. These materials are involved in numerous human activities (e.g. mining, construction, pharmacy, and agriculture), and various natural phenomenon (e.g. earthquakes, avalanches, and sedimentation) [2]. Despite the ubiquity of granular media, their mechanical behaviors remain poorly understood due to complexity, particularly in areas including jamming (solid-like state) [3], fabric evolution [4], strong nonlinearity [5], and grain breakage [6]. Recent studies have demonstrated strong links between macro-mechanical responses and underlying

microstructures and micromechanisms [7]. Exploring particle-scale behaviors and microstructure of granular media contributes to alleviating the difficulties encountered with the geotechnical constructions involving granular materials and optimizing geotechnical designs.

The analysis of microscale behaviors in granular media was significantly difficult before the advent of the X-ray micro-computed tomographic (μ CT) technique. Micromechanical characteristics such as intergrain contacts, grain orientations, porosity, etc., can be measured through three-dimensional (3D) grain reconstruction and post-processing of μ CT images [8,9]. The most used method for 3D grain reconstruction is watershed segmentation, recognized and incorporated in commercial and open-source software such as Avizo and ImageJ

^{*} Corresponding author.

E-mail address: zhenyu.yin@polyu.edu.hk (Z.-Y. YIN).

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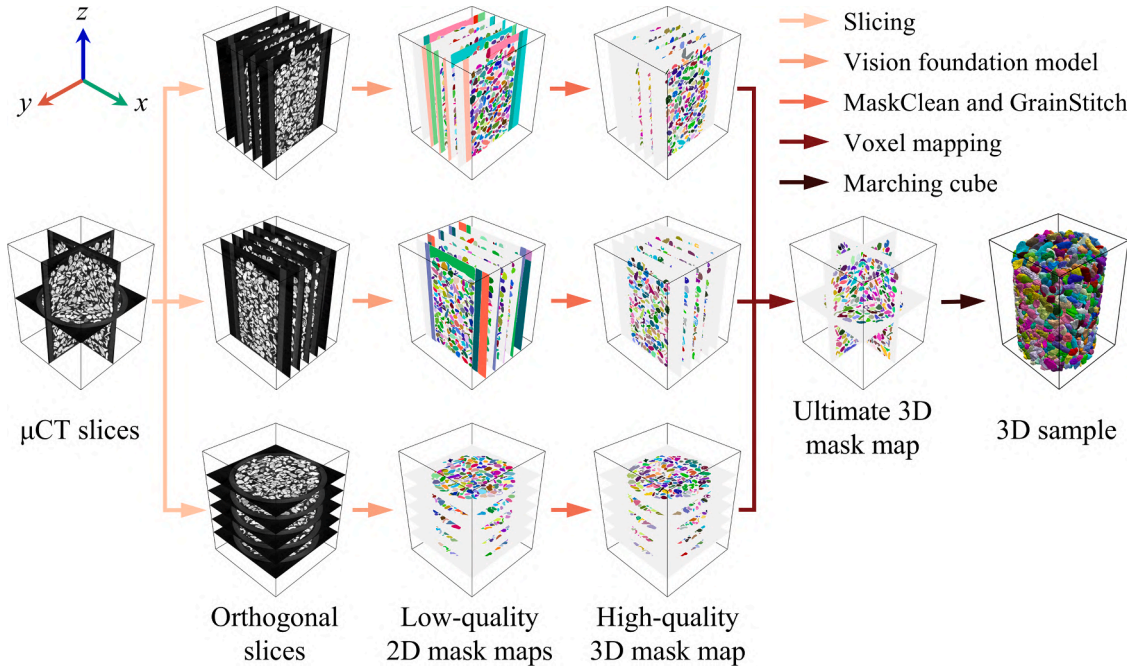


Fig. 1. Overview of GeoVision.

[10–12]. This method treats images as topographic maps and separates contacted grains by simulating water filling from the local minima to the maximum peaks of the terrain. Local minima are typically identified using the inverted Euclidean distance transform (iEDT), based on binarized images processed with the Otsu thresholding algorithm [13].

Watershed segmentation exhibits satisfactory performance with regular grains (e.g. spheres) but behaves poorly for irregular granular media due to under- and over-segmentation, such as carbonate sands. This limitation arises from that many local minima identified in iEDT are noise artifacts induced by grain shapes deviating from spherical [14]. Therefore, the H-extreme transform is proposed to alleviate this problem by measuring the difference in depth of neighboring local minima and removing those below a preset threshold. For porous grains, Kong and Fonseca [15] suggest a ‘hole-filling’ procedure before implementing H-extreme watershed, along with a ‘hole-repairing’ step following segmentation. This approach is considered state-of-the-art (SOTA) for 3D grain reconstruction of irregular porous granular media [16,17]. Despite these advancements, issues of under- and over-segmentations persist in the resulting images, necessitating considerable manual effort for correction. This limitation hinders the exploration of the complicated mechanical behaviors of granular media.

More recently, vision foundation models (VFMs) have become extremely popular in image segmentation, such as the Segment Anything Model (SAM) [18] and Seggpt [19]. These models trained on a large-scale dataset that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks are considered superior alternatives to conventional machine learning (ML) methods [20–24]. The increasing use of VFMs for two-dimensional (2D) medical image segmentation has demonstrated their feasibility and robustness [25,26]. For instance, the SAM model was used for 3D medical image segmentation and outperformed several SOTA methods [27]. The strong generalization capability of VFMs poses the potential for reconstructing arbitrary 3D granular media. However, no foundation models have been used to reconstruct 3D granular grains from μ CT images. The difficulties arise from the lack of strategy for preserving interior grain features (e.g. holes) in 2D images and the conversion from stacked 2D images to 3D images. In addition, there is no systematic evaluation of VFMs in the 2D segmentation of granular media. Therefore, it is necessary to apply VFMs to 3D grain reconstruction problems to verify feasibility and

compare them with SOTA.

To this end, this paper introduces a novel framework for automatically reconstructing 3D arbitrary granular media from μ CT images. This framework can be used to facilitate the exploration of the micro-mechanisms of granular media, thereby optimizing geotechnical designs. The novel academic contributions of this paper are summarised as follows:

- (1) The first generic framework for 3D reconstruction of arbitrary granular media utilizing VFMs is proposed. This framework allows for an automatic 3D reconstruction without training while achieving high-accuracy results.
- (2) A systematic assessment of model sizes and prompt numbers on the performance of VFMs in 3D reconstruction is presented, serving as a reference for selecting appropriate VFM.
- (3) An effective strategy for extracting high-resolution interior grain textures and a robust optimal transport (OT)-based algorithm for accurately converting stacked 2D segmentations into 3D segmentation are proposed and validated.
- (4) The proposed framework has been validated to outperform SOTA watershed segmentation by 50 % accuracy on carbonate sands and achieves 95 % accuracy on lentil samples captured in triaxial loading, demonstrating its robustness in application projects.

The paper is organized as follows: Section 2 introduces the principle of our proposed framework. Section 3 describes the experiment settings for validation and evaluation. Section 4 presents 3D visualization, accuracy analysis, inference time analysis, comparison with SOTA, and application of our method. Section 5 concludes this paper.

2. Methodology

2.1. Overview of the proposed approach

Fig. 1 presents the innovative framework ‘GeoVision’, the first generic approach developed based on the VFMs, designed to accurately reconstruct arbitrary granular media from μ CT slices. The first step is to turn μ CT slices along the x, y, and z-axes into stacked low-quality 2D mask maps by VFMs. In the mask map, the value 0 denotes the

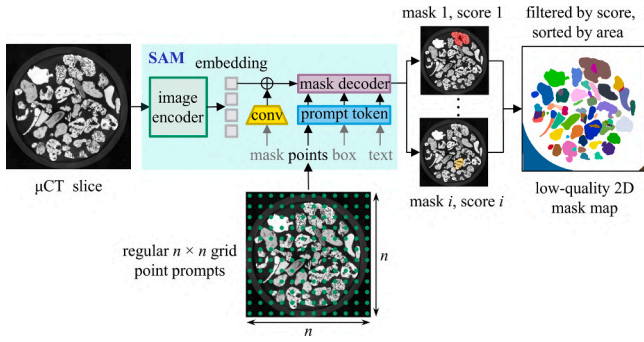


Fig. 2. Generation of 2D low-quality mask map by SAM.

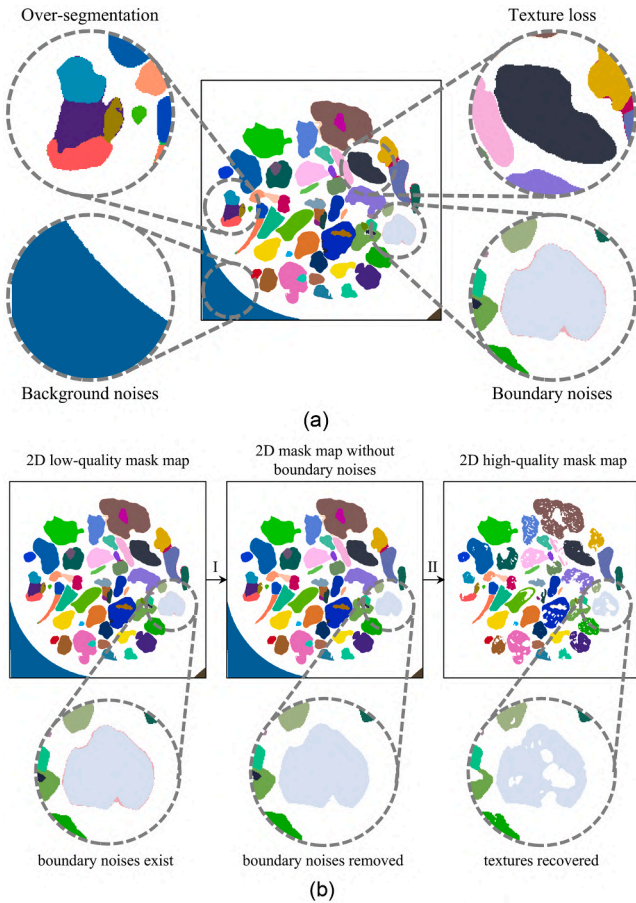


Fig. 3. MaskClean strategy: (a) existing problems; and (b) transform from 2D low-quality mask map to high-quality mask map.

background, and any value larger than 0 represents a grain. Each grain in the mask map is represented by a unique color for visualization. The proposed 'MaskClean' and 'GrainStitch' algorithms then transform 2D mask maps into high-quality 3D mask maps. The final procedure is to map three 3D mask maps obtained along three axes into the complete one and convert them into the 3D model. This method is versatile and can be straightforwardly transferred to reconstruct other materials.

2.2. Vision foundation model

The first VFM for image segmentation is the SAM, proposed by Kirillov et al. [18]. SAM is designed to segment any objects in any image with cheap input prompts (e.g., one point) without the need for additional training. Therefore, a flexible zero-shot segmentation approach is

adopted in SAM, as shown in Fig. 2. The main components of SAM are an image encoder, a prompt encoder, and a mask decoder. Image encoder transforms an input image into a downsampled embedding. The dense (masks) or sparse (points, boxes, text) prompts are distilled into embeddings by the prompt encoder [22,28]. The mask decoder outputs a confidence score for each mask according to image and prompt embeddings. Based on the training data scaling of encoders, SAM is classified into three different types: ViT-B (91 M), ViT-L (308 M), and ViT-H (636 M).

Many SAM variants or other vision foundation models have been proposed, such as HQ-SAM and FastSAM [29,30]. However, in this study, only the SAM is selected as the representative of VFMs for validation. Additionally, the regular $n \times n$ grid points are chosen as input prompts to achieve fully automatic mask generation (see Fig. 2), where n denotes the number of prompts per row or column, also called prompt size. Other prompts are also feasible but are more complicated and therefore excluded from this study [18]. For each generated mask, the model evaluates it with a score. Masks with an unsatisfactory score are filtered out, while the remaining masks are sorted by size in descending order and mapped to form a low-quality 2D mask map.

2.3. MaskClean strategy

The 2D mask map generated by the foundation model preserves the coarse boundaries of grains but is still unsatisfactory for use directly. Specifically, these masks suffer from four issues: (1) over-segmentation; (2) texture loss (e.g. interior holes); (3) background noises (unnecessary masks); and (4) boundary noises (small, isolated, or connected pixels), as illustrated in Fig. 3a. To solve these problems, the MaskClean strategy is proposed, as illustrated in Fig. 3b. The first step is to remove boundary noises, as described as follows:

$$\mathbf{M}' = \{\mathbf{m} \in \mathbf{M} | A(\mathbf{m}) \geq A_0\} \quad (1)$$

where $\mathbf{M}$ and $\mathbf{M}'$ denote the old and new 2D mask maps, respectively. $\mathbf{m}$ represents a mask map of the same size but only contains a single mask. $A(\mathbf{m})$ is the area of mask $\mathbf{m}$. A_0 is the preset threshold area, depending on the grain size, taken as 1000 pixels, which is sufficient in this study.

The second step is to recover interior textures and remove background noises by following procedures:

$$\mathbf{M}' = \{\mathbf{m} \in \mathbf{M} | H(\mathbf{m})\} \quad (2)$$

$$H(\mathbf{m}) = \begin{cases} \text{OTSU}(\mathbf{I} \odot \mathbf{m}), & \text{if } \bar{I} \geq I_0 \\ \mathbf{O}, & \text{if } \bar{I} < I_0 \end{cases} \quad (3)$$

where $\odot$ denote the Hadamard product, OTSU($\cdot$) is the Otsu segmentation [13], $\mathbf{I}$ refers to the μ CT slice, $\mathbf{O}$ represents the zero matrix, $\bar{I}$ is the average pixel value of mask $\mathbf{m}$ in $\mathbf{I}$, and I_0 denotes the threshold determined by applying the Otsu segmentation to $\mathbf{I}$. For over-segmented masks, no additional remedies are required. These over-segmented 2D masks will be repaired automatically by GrainStitch in 3D transformation, which will be discussed in the results.

2.4. GrainStitch algorithm

GrainStitch, an OT-based method, is proposed here to transform stacked 2D mask maps into 3D mask maps automatically [31,32]. Suppose there are j and k masks in two adjacent 2D mask maps along the same axis, $\mathbf{M}_i \in \mathbb{R}$ and $\mathbf{M}_{i+1} \in \mathbb{R}$, respectively. The transport plan $\mathbf{L}_{i,i+1} \in \mathbb{R}$ describes how to relabel masks in $\mathbf{M}_{i+1}$ so that masks corresponding to the same grain have the same label in two maps. In terms of OT, $\mathbf{M}_i$ and $\mathbf{M}_{i+1}$ can be regarded as two uniform discrete probability distributions:

$$\mathbf{M}_i(p) = \frac{1}{j} \quad (4)$$

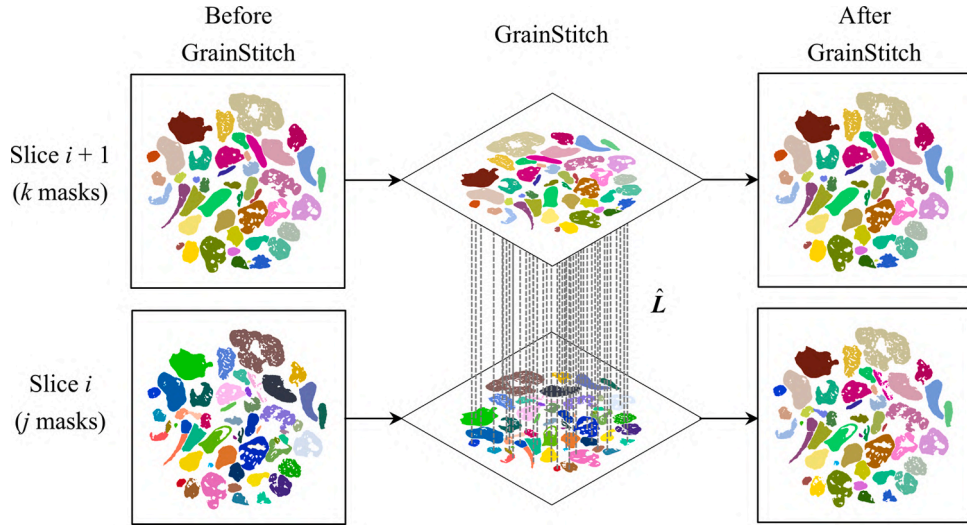
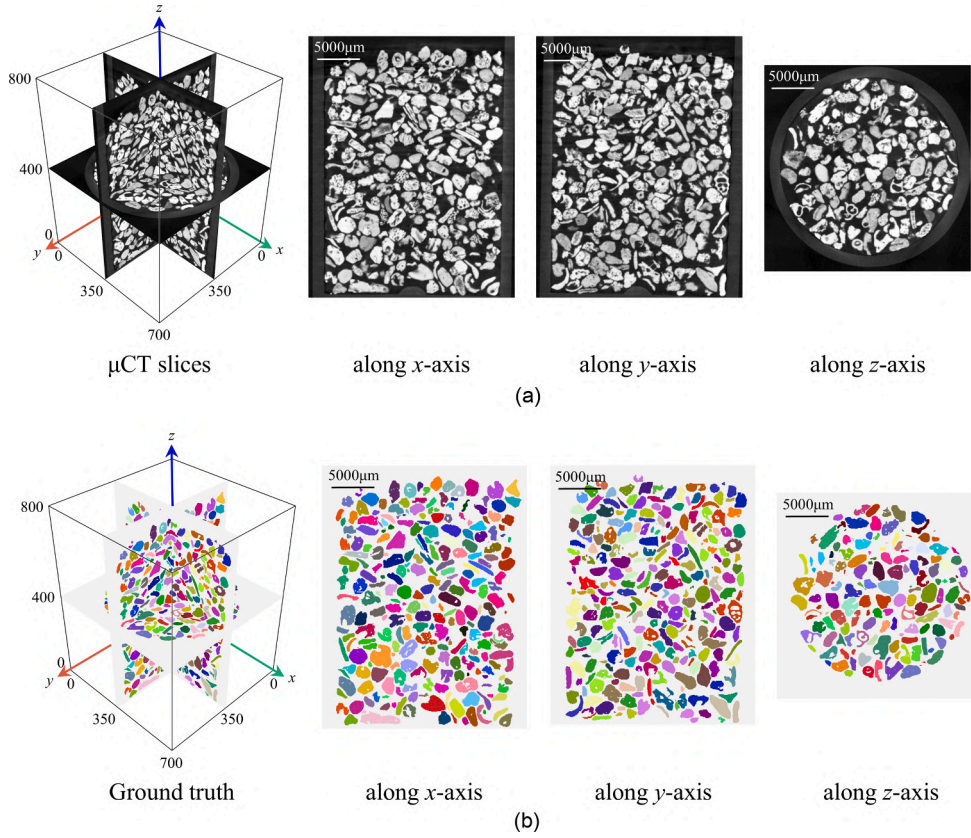


Fig. 4. Explanation of GrainStitch algorithm.

Fig. 5. Dataset CS1: (a) μ CT slices; and (b) ground truth.

$$\mathbf{M}_{i+1}(q) = \frac{1}{k} \quad (5)$$

where p and q represent the mask in $\mathbf{M}_i$ and $\mathbf{M}_{i+1}$, respectively. The optimal $\mathbf{L}_{i,i+1}$ is solved by Kantorovich OT, which is heavily determined by the cost function [33]:

$$\hat{\mathbf{L}}_{i,i+1} = \arg \min_{\mathbf{L} \in \Gamma(\mathbf{M}_i, \mathbf{M}_{i+1})} \sum_{p,q} C_{p,q} L_{p,q} \quad (6)$$

$$\Gamma(\mathbf{M}_i, \mathbf{M}_{i+1}) = \left\{ \mathbf{L} \in \mathbb{R} : \sum_q L_{p,q} = \mathbf{M}_i, \sum_p L_{p,q} = \mathbf{M}_{i+1} \right\} \quad (7)$$

where $\hat{\mathbf{L}}_{i,i+1}$ is the optimal $\mathbf{L}_{i,i+1}$, $L_{p,q}$ represents the matching between masks p and q , and $C_{p,q}$ is the corresponding cost function, defined as follows:

$$C_{p,q} = 1 - \max \left\{ \frac{I(p,q)}{U(p,q)}, \frac{I(p,q)}{N(p)}, \frac{I(p,q)}{N(q)} \right\} \quad (8)$$

where $\max(\cdot)$ denotes the maximum value, $I(p, q)$ and $U(p, q)$ are the

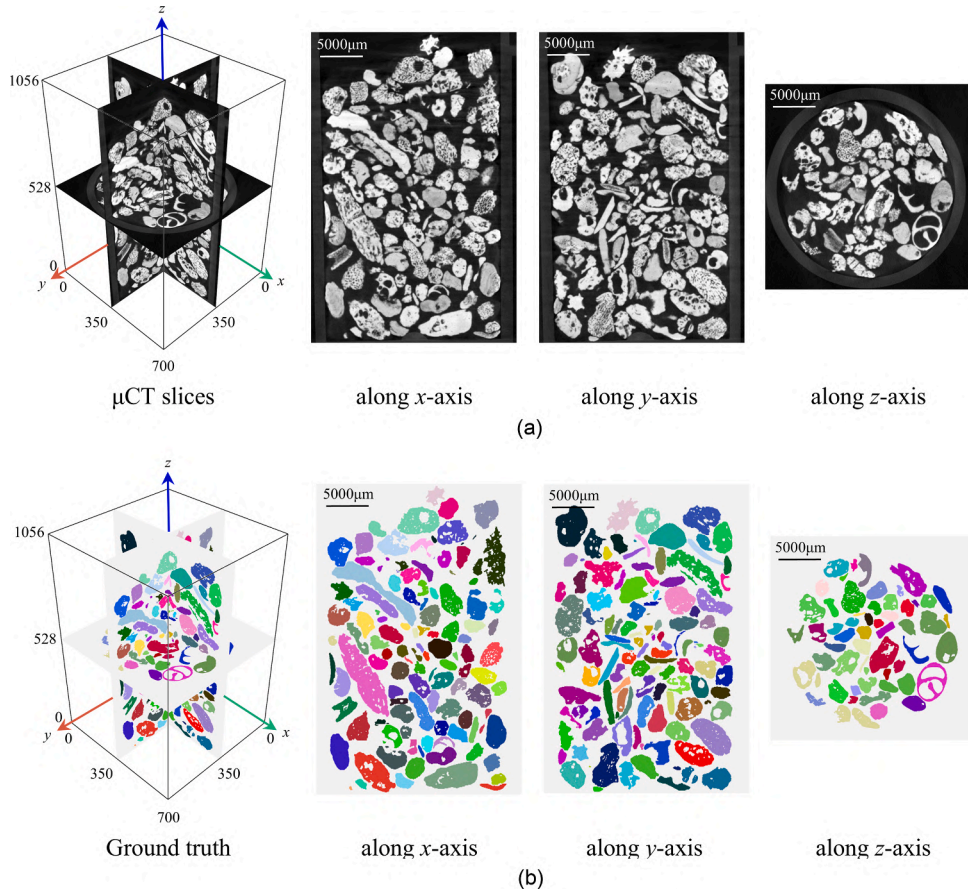


Fig. 6. Dataset CS2: (a) μ CT slices; and (b) ground truth.

Table 1
Comparative tests across datasets CS1 and CS2.

Dataset	Model	Parameters	Dataset	Model	Parameters
CS1	GeoVision (SAM-ViT-B)	$n = 8, 16, 24, 32, 40$	CS2	GeoVision (SAM-ViT-B)	$n = 8, 16, 24, 32, 40$
CS1	GeoVision (SAM-ViT-L)	$n = 8, 16, 24, 32, 40$	CS2	GeoVision (SAM-ViT-L)	$n = 8, 16, 24, 32, 40$
CS1	GeoVision (SAM-ViT-H)	$n = 8, 16, 24, 32, 40$	CS2	GeoVision (SAM-ViT-H)	$n = 8, 16, 24, 32, 40$

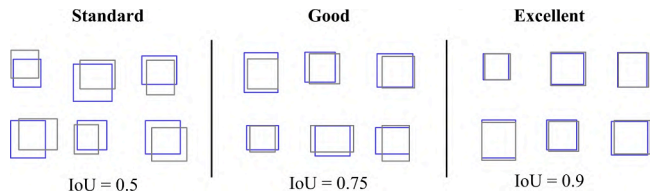


Fig. 7. Explanation of different IoU matching thresholds.

number of pixels in the intersection or union of mask p in $\mathbf{M}_i$ and mask q in $\mathbf{M}_{i+1}$, respectively. The ratio between $I(p, q)$ and $U(p, q)$ is the so-called Intersection over Union (IoU). IoU provides a quantitative measure of comparing the similarity between two arbitrary shapes [34]. $N(p)$ and $N(q)$ represent the number of pixels of mask p in $\mathbf{M}_i$ and mask q in $\mathbf{M}_{i+1}$, respectively. This novel cost function incorporates IoU and the ratio between the intersection area and two mask areas. In this study, the

Sinkhorn algorithm is adopted to solve Kantorovich OT with the entropic regularization term equal to 0.15 [35].

Note that $\hat{\mathbf{L}}_{i,i+1}$ may not be a complete solution. For a given mask q in $\mathbf{M}_{i+1}$, more than one grain in $\mathbf{M}_i$ could be appropriate to match with q . Nevertheless, only the mask in $\mathbf{M}_i$ with the lowest cost is selected to be matched with the mask q in $\mathbf{M}_{i+1}$ if merely based on $\hat{\mathbf{L}}_{i,i+1}$. Therefore, an additional OT $\hat{\mathbf{L}}_{i+1,i} \in \mathbb{R}$ computed inversely for relabeling masks in $\mathbf{M}_i$ based on $\mathbf{M}_{i+1}$ is needed. Lastly, the ultimate OT $\hat{\mathbf{L}}$ for updating masks in $\mathbf{M}_i$ and $\mathbf{M}_{i+1}$ is the union of $\hat{\mathbf{L}}_{i,i+1}$ and $\hat{\mathbf{L}}_{i+1,i}$, as shown in Fig. 4:

$$\hat{\mathbf{L}} = \hat{\mathbf{L}}_{i,i+1} \cup \hat{\mathbf{L}}_{i+1,i} \quad (9)$$

For each mask q in $\mathbf{M}_{i+1}$, the optimal mask $\hat{p}$ in $\mathbf{M}_i$ for matching is the one with the least cost in $\hat{\mathbf{L}}$. Notably, if the value of $\hat{p}$ in $\mathbf{M}_i$ is 0, the mask q in $\mathbf{M}_{i+1}$ is assumed to be a new mask, assigned with a new label $j + 1$. Additionally, if the mask p in $\mathbf{M}_i$ is determined to belong to a mask $\hat{q}$ in $\mathbf{M}_{i+1}$, all masks in the processed 2D mask maps (e.g., mask map $\mathbf{M}_{i-1}$) that have the same value as p are relabeled to the mask $\hat{q}$. The GrainStitch algorithm is implemented between adjacent 2D mask maps along the same axis sequentially until all 2D mask maps are relabeled. Finally, the three 2D stacked mask maps along the x , y , and z -axes are transformed into three corresponding 3D mask maps by GrainStitch.

2.5. Voxel mapping

Suppose the three 3D mask maps (or 3D voxels) stitched along the x , y , and z -axes are $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$, respectively. The ultimate 3D mask map $\mathbf{T}$ is obtained as follows:

$$\mathbf{T} = \mathbf{S} \circ \mathbf{B}(\mathbf{X}) \circ \mathbf{B}(\mathbf{Y}) \circ \mathbf{B}(\mathbf{Z}) \quad (10)$$

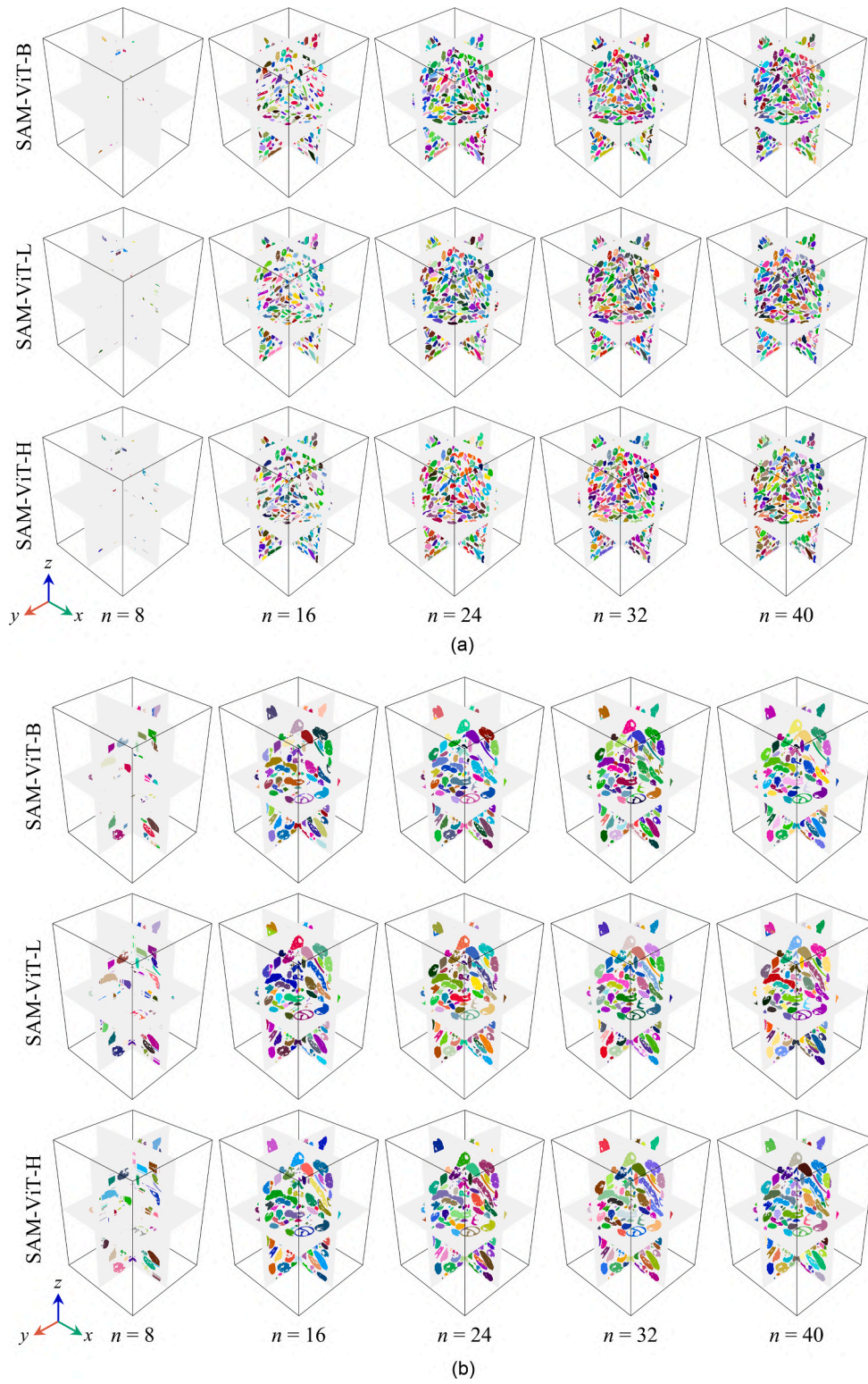


Fig. 8. 3D mask maps obtained by GeoVision: (a) CS1; and (b) CS2.

where S is the mapped 3D mask map by assigning voxel with the same label values in X , Y , and Z , and $B(\cdot)$ denotes the binarization. Subsequently, the 3D grains are reconstructed by the marching cube algorithm based on the final 3D mask map T and smoothed by Gaussian filtering with a kernel size of 3×3 [36,37].

3. Experiment

3.1. Materials, apparatus, and benchmark

Two complex uncemented shelly carbonate sand samples from the South China Sea were tested: one comprised grains ranging from 1 to 2 mm (denoted as dataset CS1), and the other consisted of grains with grain sizes between 2 and 5 mm (denoted as dataset CS2). The

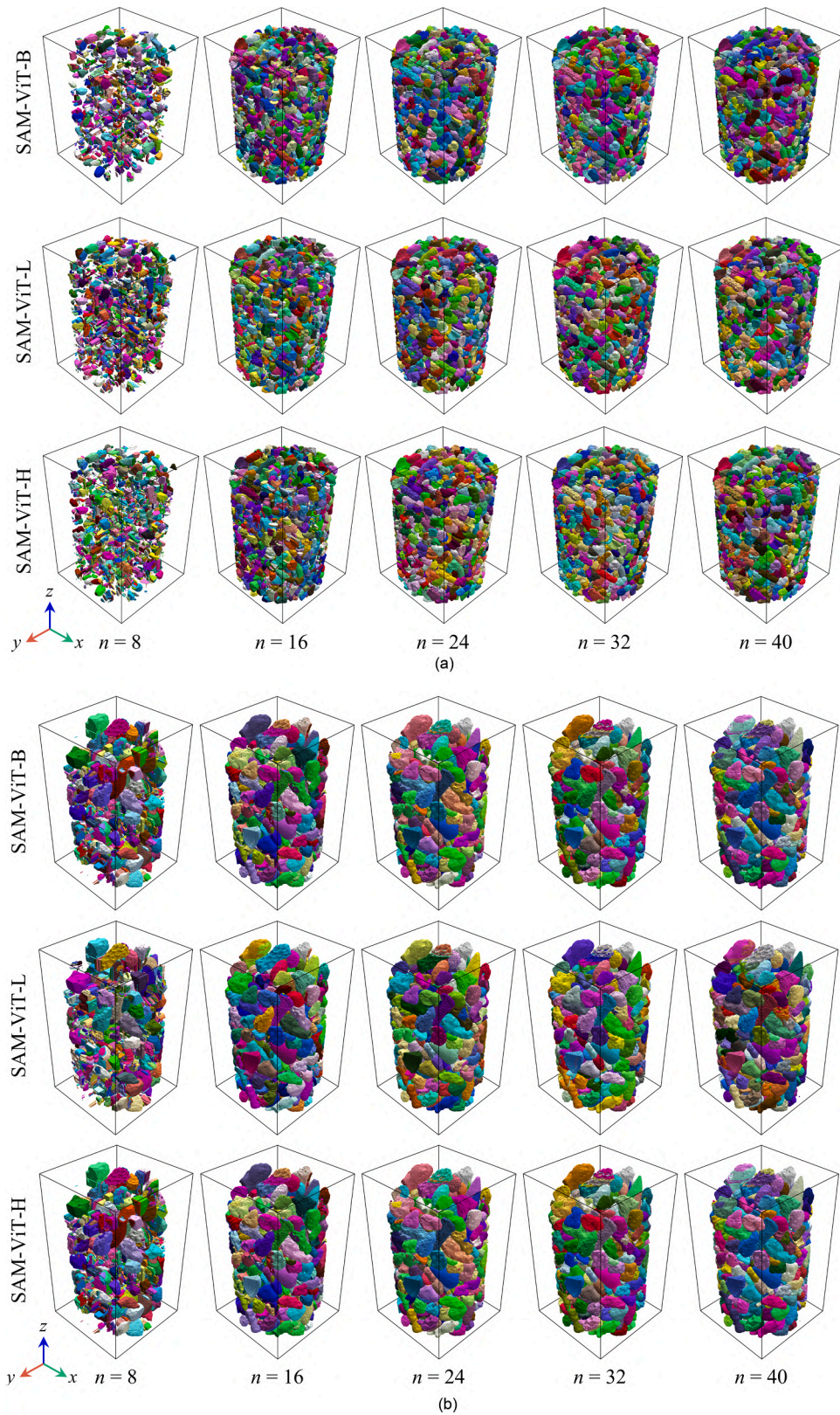


Fig. 9. 3D samples obtained by GeoVision: (a) CS1; and (b) CS2.

mechanical behavior of these carbonate sands had been previously investigated [38–40]. The μ CT images of these two samples were obtained by the Nikon XT H225ST 2X μ CT system. The major components of this μ CT system include an X-ray beam tub, a high-voltage generator, a sample manipulator, and a flat panel detector. The generator powers

the beam tube to produce X-rays, which pass through the sample and cast a shadow image on the detector. The shadow image is a single-angle projection of the sample. For high-quality μ CT images, 3000 projections were taken for each scan as the sample rotated 360°. During X-ray scanning, objects absorb varying energy levels depending on their

Table 2

Model performance of GeoVision in 3D Reconstruction of CS1.

Model type	n	IoU = 0.5			IoU = 0.75			IoU = 0.9		
		Precision	Recall	AP	Precision	Recall	AP	Precision	Recall	AP
SAM-ViT-B	8	0.212	0.056	0.046	0.061	0.016	0.013	0.013	0.003	0.003
	16	0.402	0.526	0.295	0.232	0.304	0.151	0.077	0.101	0.046
	24	0.782	0.866	0.698	0.698	0.773	0.579	0.492	0.545	0.349
	32	0.878	0.912	0.809	0.824	0.856	0.724	0.701	0.729	0.556
	40	0.923	0.935	0.868	0.879	0.890	0.792	0.799	0.810	0.673
SAM-ViT-L	8	0.168	0.065	0.049	0.054	0.021	0.015	0.012	0.005	0.003
	16	0.428	0.623	0.340	0.269	0.392	0.190	0.100	0.146	0.063
	24	0.827	0.881	0.744	0.747	0.795	0.626	0.569	0.606	0.415
	32	0.905	0.913	0.833	0.852	0.859	0.747	0.725	0.731	0.572
	40	0.905	0.913	0.833	0.849	0.857	0.743	0.740	0.747	0.592
SAM-ViT-H	8	0.172	0.064	0.049	0.057	0.021	0.016	0.009	0.003	0.002
	16	0.421	0.602	0.330	0.261	0.373	0.182	0.093	0.133	0.058
	24	0.830	0.883	0.748	0.750	0.798	0.630	0.552	0.588	0.398
	32	0.894	0.909	0.821	0.834	0.848	0.725	0.711	0.722	0.558
	40	0.894	0.915	0.826	0.836	0.855	0.732	0.732	0.748	0.587

Table 3

Model performance of GeoVision in 3D Reconstruction of CS2.

Model type	n	IoU = 0.5			IoU = 0.75			IoU = 0.9		
		Precision	Recall	AP	Precision	Recall	AP	Precision	Recall	AP
SAM-ViT-B	8	0.081	0.162	0.057	0.028	0.056	0.019	0.004	0.008	0.003
	16	0.426	0.766	0.377	0.344	0.618	0.283	0.198	0.356	0.146
	24	0.700	0.920	0.660	0.644	0.846	0.576	0.511	0.672	0.409
	32	0.777	0.934	0.737	0.729	0.876	0.661	0.657	0.790	0.559
	40	0.768	0.934	0.729	0.717	0.872	0.649	0.663	0.806	0.572
SAM-ViT-L	8	0.092	0.214	0.069	0.040	0.094	0.029	0.007	0.016	0.005
	16	0.484	0.824	0.438	0.410	0.698	0.348	0.258	0.440	0.194
	24	0.773	0.932	0.732	0.721	0.870	0.651	0.625	0.754	0.519
	32	0.793	0.932	0.749	0.748	0.880	0.679	0.687	0.808	0.591
	40	0.773	0.928	0.730	0.723	0.868	0.652	0.663	0.796	0.567
SAM-ViT-H	8	0.089	0.212	0.067	0.036	0.086	0.026	0.005	0.012	0.004
	16	0.455	0.820	0.414	0.380	0.684	0.323	0.222	0.400	0.167
	24	0.763	0.936	0.726	0.698	0.856	0.625	0.608	0.746	0.504
	32	0.784	0.938	0.746	0.734	0.878	0.666	0.662	0.792	0.564
	40	0.748	0.928	0.707	0.698	0.866	0.630	0.632	0.784	0.538

material composition and density, with denser materials absorbing more energy. This absorption difference is reflected in the intensity values of the voxels, allowing for feature differentiation based on contrast in the intensity levels.

The μ CT slices of CS1 and CS2 are presented in Fig. 5a and Fig. 6a, respectively. It is evident that the carbonate sands exhibit highly irregular shapes and significant porosity. These complex morphological characteristics make carbonate sands an ideal material for testing the feasibility and robustness of the GeoVision framework. The spatial resolution of these μ CT images is $35 \times 35 \times 35 \mu\text{m}$. The dimensions of CS1 and CS2 are $700 \times 700 \times 800$ and $700 \times 700 \times 1056$ voxels, respectively. There are 2123 and 501 grains in CS1 and CS2, respectively. The corresponding ground truth of CS1 and CS2, represented in the form of 3D mask maps, are illustrated in Fig. 5b and Fig. 6b, respectively. These ground truths are determined by manual labeling and will be open source.

To evaluate the performance of GeoVision in 3D grain reconstruction, a series of comparative tests were conducted and summarized in Table 1. Three SAM models (SAM-ViT-B, SAM-ViT-L, and SAM-ViT-H) were used to evaluate the effect of model size on performance. The variable n was set to 8, 16, 24, 32, and 40 for analyzing the influence of prompt numbers on 3D reconstruction. For comparison, the SOTA H-extreme watershed with hole-filling and hole-repairing is adopted as the benchmark in this study [10,15]. All computations in this study are performed on a PC with a 13th Gen Inter(R) Core i5-13600KF CPU, 64 GB RAM, and NVIDIA GeForce RTX 4090 GPU.

3.2. Evaluation metrics

To evaluate the reconstruction accuracy qualitatively, two types of segmentation errors are used: over- and under-segmentation [41]. Over-segmentation refers to a complete grain being segmented into multiple pieces while under-segmentation indicates that more than one grain remains connected. Alternatively, the following standard metrics are adopted to evaluate the reconstruction accuracy quantitatively: Precision, Recall, and Average Precision (AP). The three metrics are defined as follows [42]:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (11)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (12)$$

$$\text{AP} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \quad (13)$$

where TP is the number of matched masks, FP is the number of unmatched masks in the predicted result, and FN is the number of unmatched masks in the ground truth. In this study, the ground truth and candidate mask are regarded as matched if the IoU between the two is greater than a threshold. The standard definition of IoU is used, denoting the ratio of the intersecting volume between the bounding boxes of the ground truth and candidate mask to the total volumes encompassed by both [34]. Fig. 7 presents schematic diagrams illustrating various IoU matching thresholds. An IoU threshold of 0.5 is commonly adopted as

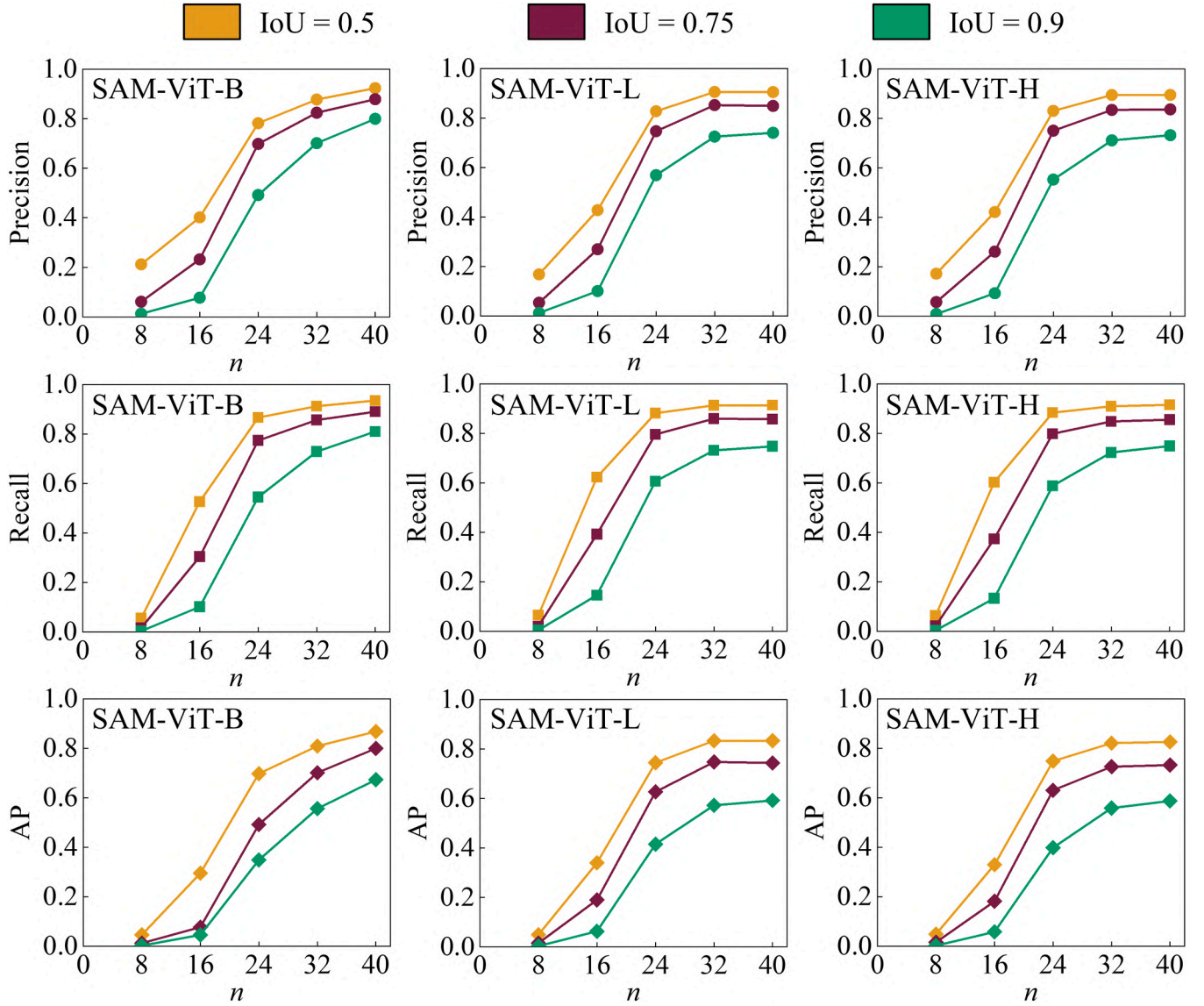


Fig. 10. Influence of the model sizes and prompt sizes on the accuracy of GeoVision at different IoU thresholds for reconstructing CS1.

the standard for evaluation [31]. Segmentation quality is enhanced with increasing IoU thresholds, where 0.75 signifies good segmentation and 0.9 denotes excellent segmentation. The IoU metric serves as a quantitative measure of model performance in accurately segmenting objects within an image, thereby establishing its value as a tool for assessing segmentation efficacy. If the IoU matching threshold is set to 0.75, Precision, Recall, and AP can be denoted as $\text{Precision}_{0.75}$, $\text{Recall}_{0.75}$, and $\text{AP}_{0.75}$, respectively.

4. Results

4.1. 3D visualization

The 3D mask maps and samples reconstructed by GeoVision for both CS1 and CS2 are shown in Fig. 8 and Fig. 9, giving a glimpse of visual inspection, respectively. In both cases, the majority of grains are not reconstructed successfully at prompt size $n = 8$, demonstrated by nearly blank slices shown in Fig. 8 and incomplete grain morphologies illustrated in Fig. 9. These poor results of GeoVision are attributed to an insufficient number of input prompts. Detailedly, the vision foundation model is not fed with sufficient prompts to position all grains in 2D mask extraction, resulting in 3D grain loss. The simplest solution for this

problem is to increase the prompt numbers. For instance, at $n = 16$, there is a noticeable improvement in the number of reconstructed grains, with the overall outline of the sample becoming visible. However, both over-segmentation and under-segmentation are still apparent, indicating that $n = 16$ remains insufficient for both datasets.

Satisfactory results are first observed at $n = 24$ for both CS1 and CS2, where grain morphology and internal textures are well-preserved, and segmentation errors are inapparent. These findings highlight the robustness of GeoVision in reconstructing 3D grains from μCT slices. Notably, visual inspection reveals no significant differences in reconstructions among $n = 24, 32$, and 40 , suggesting that reconstruction accuracy will not consistently improve with increasing n . Once the number of input prompts is sufficient to identify all grains, further increasing n does not significantly enhance reconstruction accuracy. Additionally, when the number of prompts is sufficient, no substantial differences are observed in the reconstruction results among SAM-ViT-B, SAM-ViT-L, and SAM-ViT-H. This indicates that the quantity of prompts is more critical than model size in determining reconstruction performance.

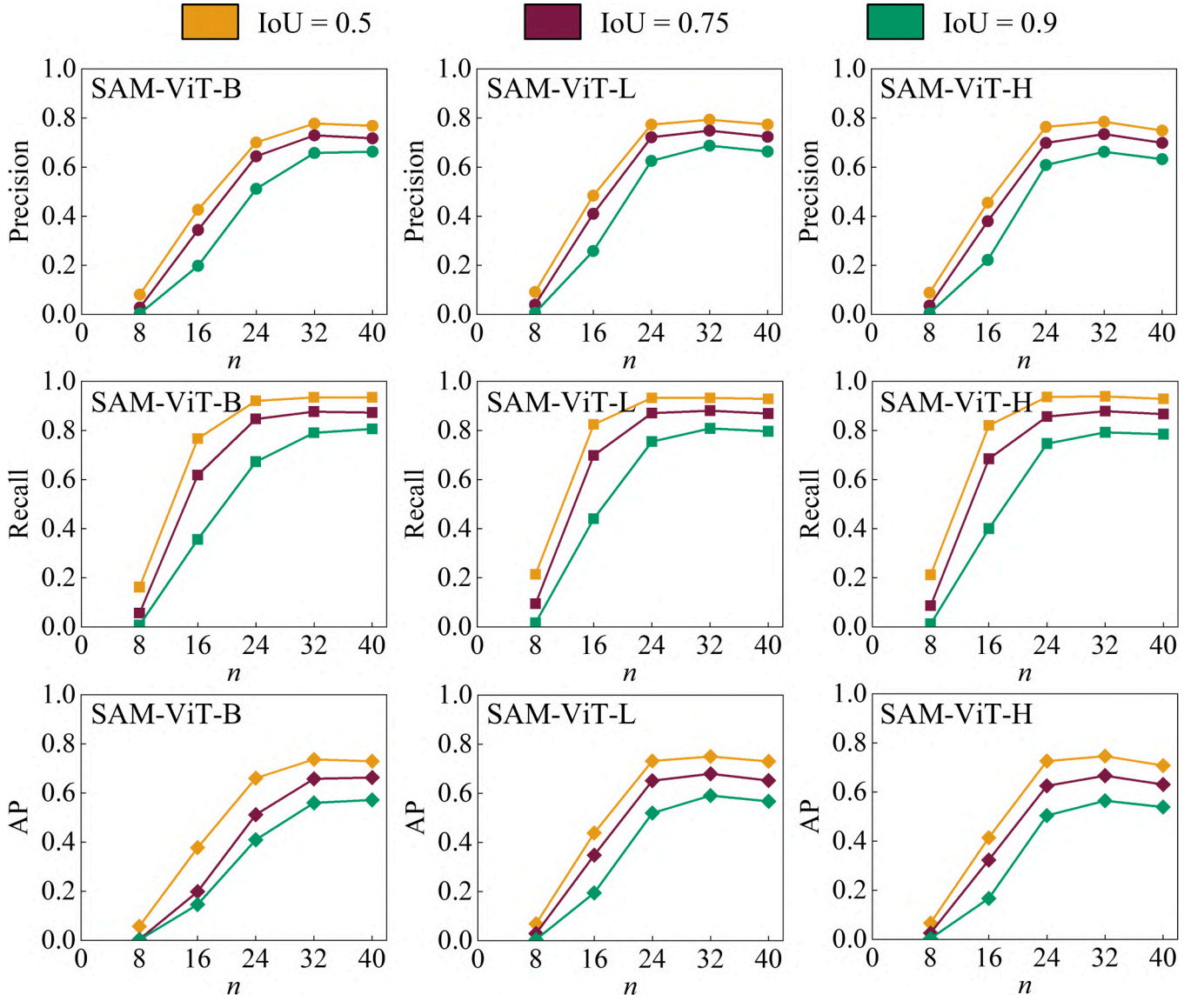


Fig. 11. Influence of the model sizes and prompt sizes on the accuracy of GeoVision at different IoU thresholds for reconstructing CS2.

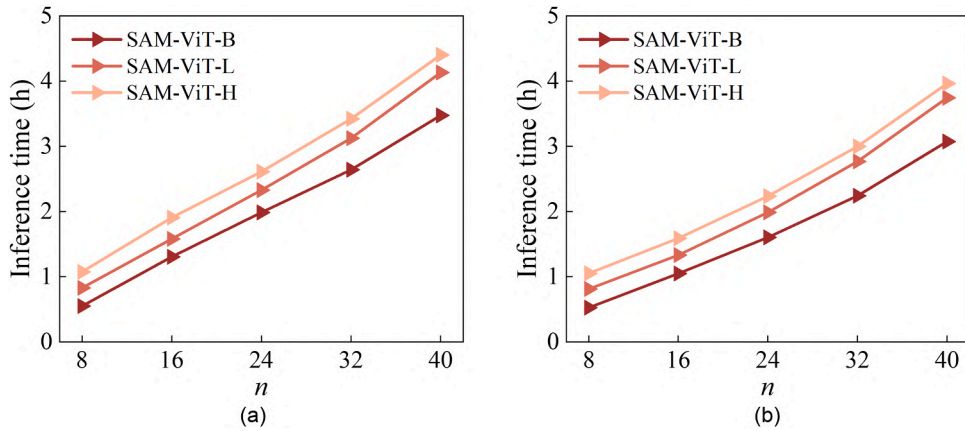


Fig. 12. Relationships between the inference time and prompt size n : (a) CS1; and (b) CS2.

4.2. Reconstruction accuracy analysis

The precision, recall, and AP values evaluated by three IoU matching

thresholds (0.5, 0.75, and 0.9) of GeoVision for both CS1 and CS2 are presented in Table 2 and Table 3, respectively. The best results in a column are marked in bold font. For CS1, the best result is obtained by

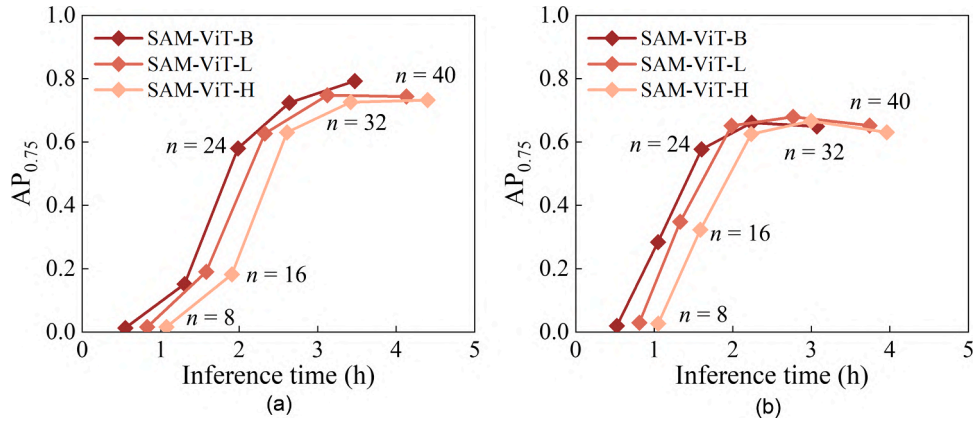


Fig. 13. Relationships between the inference time and $AP_{0.75}$: (a) CS1; and (b) CS2.

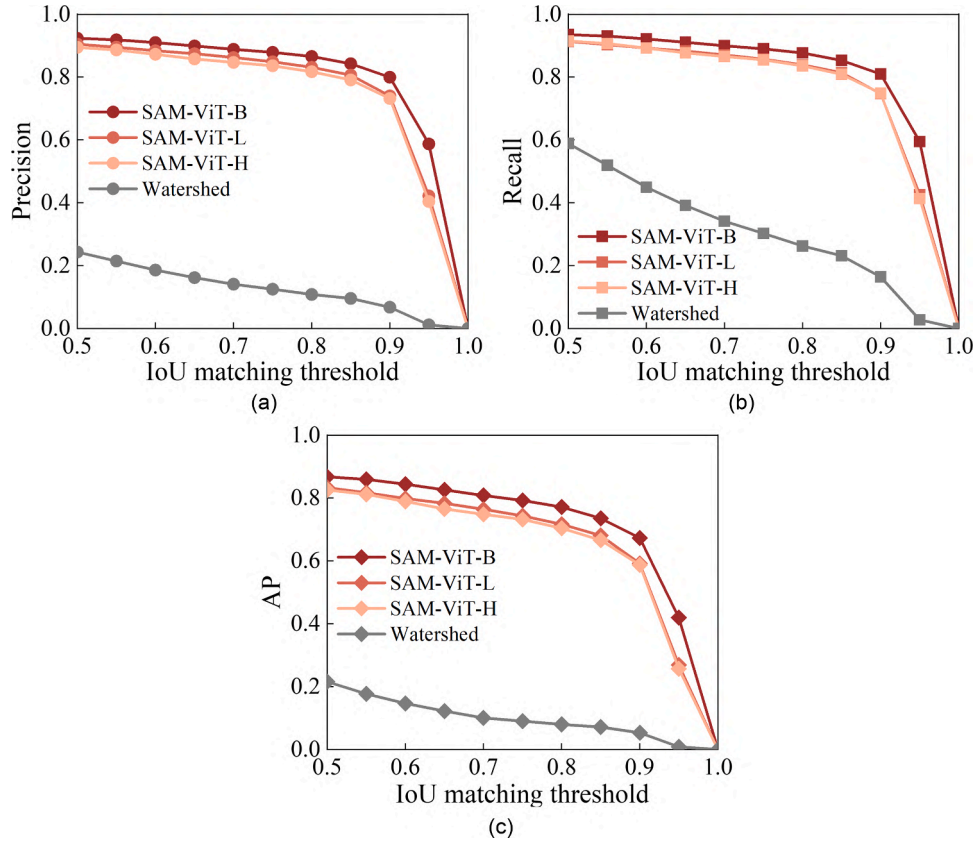


Fig. 14. Comparison of evaluation metrics in reconstructing CS1 between GeoVision and watershed at different IoU matching thresholds (from 0.5 to 1 in increments of 0.05: (a) precision; (b) recall; and (c) AP.

model SAM-ViT-B with $n = 40$, dominating all metrics at different IoU matching thresholds. The precision, recall, and AP values reach 0.923, 0.935, and 0.868, respectively, when the IoU matching threshold equals 0.5. Even for the excellent IoU matching threshold ($\text{IoU} = 0.9$), the precision, recall, and AP values still arrive at 0.799, 0.810, and 0.673, respectively. These results are acceptable given the extreme difficulty in reconstructing carbonate sands. For CS2, model SAM-ViT-B still behaves the best but achieves $n = 32$, with much fewer prompts than that of CS1. This result indicates that the optimal prompt number is related to the grain numbers. SAM-ViT-B outperforms across nearly all metrics, except for recall at $\text{IoU} = 0.5$, where it slightly lags behind SAM-ViT-H with $n = 32$. At $\text{IoU} = 0.9$, SAM-ViT-B achieves precision, recall, and AP values of 0.687, 0.808, and 0.591, respectively. The overall performance

for CS2 is slightly lower than that for CS1, likely due to the larger grain sizes in CS2, which are more prone to over-segmentation.

The influence of the model sizes and prompt sizes n on the reconstruction accuracy of GeoVision at different IoU matching thresholds (0.5, 0.75, and 0.9) is explored in Fig. 10 and Fig. 11 for both CS1 and CS2, respectively. For CS1, all metrics improve with increasing n across all IoU matching thresholds. Notably, there is a sharp rise in accuracy when $n \leq 24$, followed by a more gradual increase as n continues to grow. These trends align with the visual assessments in Fig. 8, indicating that the influence of increasing prompts on accuracy is limited once sufficient prompts are input. The metric values are lower at higher IoU thresholds compared to lower ones, which is expected, as more precise matching is inherently more challenging. Interestingly, precision, recall,

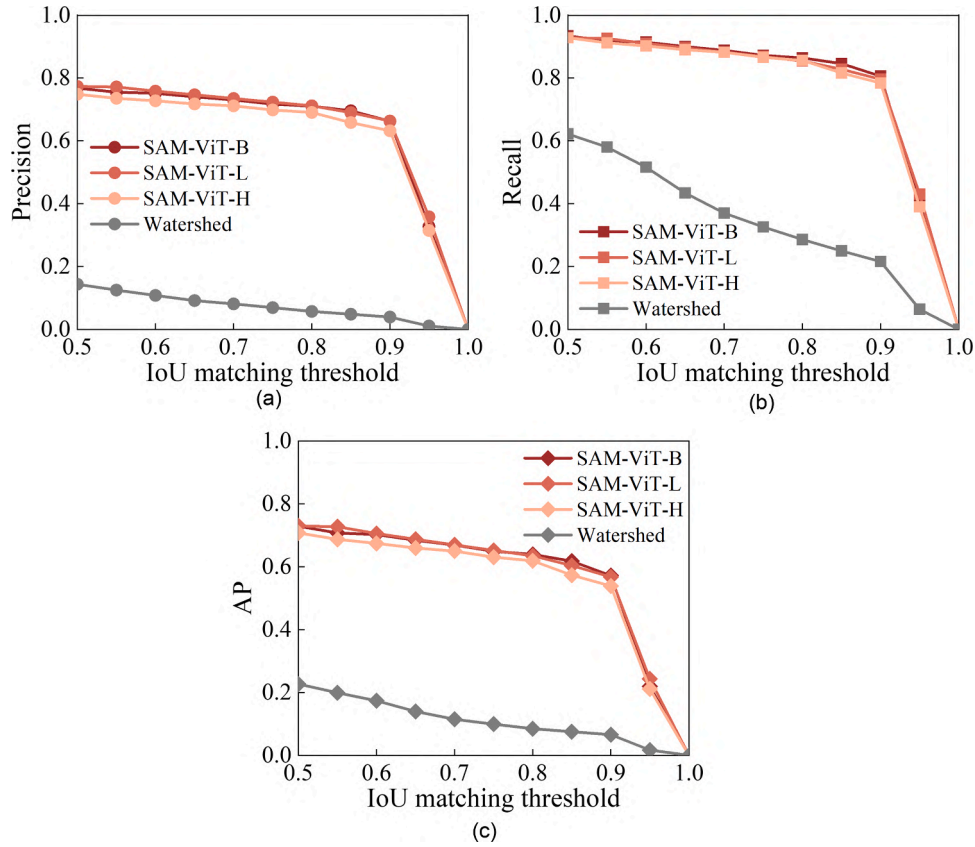


Fig. 15. Comparison of evaluation metrics in reconstructing CS2 between GeoVision and watershed at different IoU matching thresholds (from 0.5 to 1 in increments of 0.05): (a) precision; (b) recall; and (c) AP.

and AP values for the three models are similar when $n \geq 32$, consistent with observations in Fig. 8 and Fig. 9. The general performance of SAM-ViT-L and SAM-ViT-H are assumed to be superior to that of SAM-ViT-B according to tests conducted on the COCO dataset [18]. However, the optimal performance is achieved by SAM-ViT-B, indicating that the 3D grain reconstruction accuracy is not significantly determined by model size or general performance. Instead, factors such as the number of prompts are relatively more critical to reconstructions.

For CS2, a similar trend of increasing metrics with n is observed when n is less than 24. However, unlike CS1, the metrics for CS2 either plateau or decrease when n exceeds 24. For example, the $AP_{0.75}$ value at $n = 40$ of model SAM-ViT-H is smaller than that at $n = 32$. A possible explanation is that the excessive number of prompts, combined with the strong mask extraction capabilities of the VFM, leads to over-segmentation. Generally, a model fed with more prompts has a stronger ability to segment all grains, but it is also more prone to the issue of identifying the same grains multiple times, resulting in over-segmentation. These results highlight the importance of selecting the appropriate VFM and number of prompts again, as both are critical to reconstruction accuracy.

4.3. Inference time analysis

The inference time refers to the total time for VFM to extract 2D masks from μ CT slices along the x , y , and z -axes, the most prolonged compared to other procedures [18]. The time allocated to MaskClean and GrainStitch algorithms constitutes only less than 10 % of the total duration and remains relatively constant, averaging approximately 20 minutes throughout this study. Consequently, the incorporation of MaskClean and GrainStitch does not increase computational complexity; rather, it ensures the accuracy of the reconstruction. To enhance computational efficiency, it is crucial to analyze the trade-off between

inference time and reconstruction accuracy to optimize overall performance. Fig. 12 illustrates the relationships between inference time and prompt size n for the three SAM models on both CS1 and CS2 datasets. The inference time increases linearly with n across all cases. Model SAM-ViT-H is the slowest, followed by SAM-ViT-L and SAM-ViT-B, highlighting the impact of model size on inference time. This disparity in performance may be due to differences in model architecture. Replacing the current image encoder with a lighter one may reduce the inference time. The average inference time for reconstructing CS1 and CS2 at $n = 40$ for using the three models is 4 and 3.6 hours, respectively. Although these durations are relatively lengthy, the times remain within an acceptable range, ensuring satisfactory results.

Fig. 13 depicts the correlation between $AP_{0.75}$ and inference time at different n for both CS1 and CS2. The SAM-ViT-B model demonstrates the fastest performance compared to other models while achieving comparable accuracy levels. For both cases, the $AP_{0.75}$ increases with inference time before reaching a plateau. In the case of CS1, the optimal balance between accuracy and inference time is achieved at $n = 32$, where $AP_{0.75}$ approaches 0.8 and the inference time is approximately 3 hours. Similarly, the optimal inference time with satisfactory accuracy is 2 hours, achieved at $n = 24$. An excessively large n may not enhance accuracy; instead, it can increase inference time and even lead to over-segmentation. Conversely, a too-small n may fail to adequately position all grains, resulting in relatively low accuracy. The optimal n for reconstruction can be determined by a rough test and fine-tuning. For example, a size of 32×32 regular grid points can be used as an initial trial parameter. If the results prove satisfactory, a reduction in the value of n may be considered to reduce processing time, otherwise consider increasing n until reaching a satisfactory balance. In addition to optimizing the number of prompts, reducing the image size by removing unnecessary information (e.g. background) also contributes to improved efficiency.

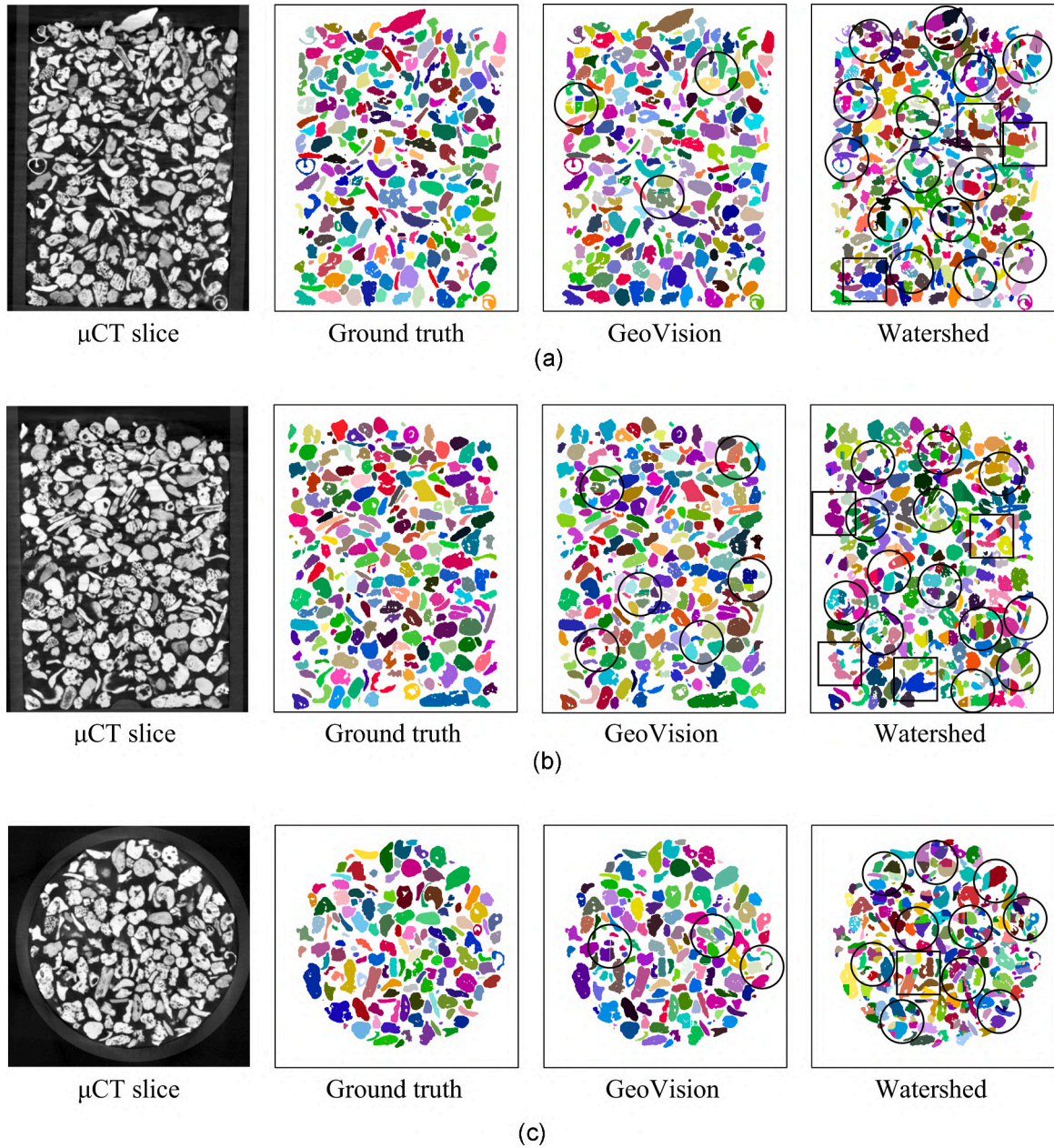


Fig. 16. 2D visualizations of representative slices along the x, y, and z-axis for CS1: (a) $x = 272$, (b) $y = 355$, and (c) $z = 701$.

4.4. Comparison with benchmark

The best performance of the watershed method is used here for comparison. Fig. 14 shows the variation of precision, recall, and AP values for reconstructing CS1 at different IoU matching thresholds (from 0.5 to 1 in increments of 0.05) by GeoVision (utilizing SAM-ViT-B, SAM-ViT-L, and SAM-ViT-H at $n = 40$) and watershed. The precision, recall, and AP values at IoU = 0.5 of GeoVision are approximately 0.9, 0.9, and 0.8, respectively. In contrast, the Precision_{0.5}, Recall_{0.5}, and AP_{0.5} values of the watershed are only around 0.2, 0.6, and 0.2, respectively. The metrics for GeoVision exhibit a slight decrease until IoU = 0.9, followed by a sharp decline to 0. This sharp drop to 0 is expected, as it is nearly impossible for two masks to be matched without any voxel discrepancies. The precision, recall, and AP values for the watershed method consistently decrease with increasing IoU. It is evident that GeoVision significantly outperforms the watershed method in terms of precision, recall, and AP across all cases. Even the worst results obtained by GeoVision exceed those of the watershed method by nearly 70 %, 50 %, and

60 % in precision, recall, and AP values, respectively. Notably, the total computational time for SAM-ViT-B, SAM-ViT-L, SAM-ViT-H, and watershed are 12,499 s, 14,867 s, 15,842 s, and 200 s, respectively. The watershed is much faster but behaves terribly in accuracy.

Fig. 15 depicts the variation of precision, recall, and AP values for reconstructing CS2 at different IoU matching thresholds (from 0.5 to 1 in increments of 0.05) by GeoVision (utilizing SAM-ViT-B, SAM-ViT-L, and SAM-ViT-H at $n = 40$) and watershed. Similar trends are observed, with metrics for both GeoVision and the watershed method experiencing a slight decrease when IoU is less than 0.9 before dropping to 0. The precision, recall, and AP values of GeoVision are shown to significantly surpass those of the watershed method once again. These results demonstrate that GeoVision is an effective framework for reconstructing complex granular media. The watershed method performs poorly due to its inherent limitation in eliminating unnecessary local minima based solely on parameter selection. This limitation prevents the watershed method from achieving accurate segmentation of all irregular shapes and may lead to significant over-segmentation. Conversely, GeoVision is

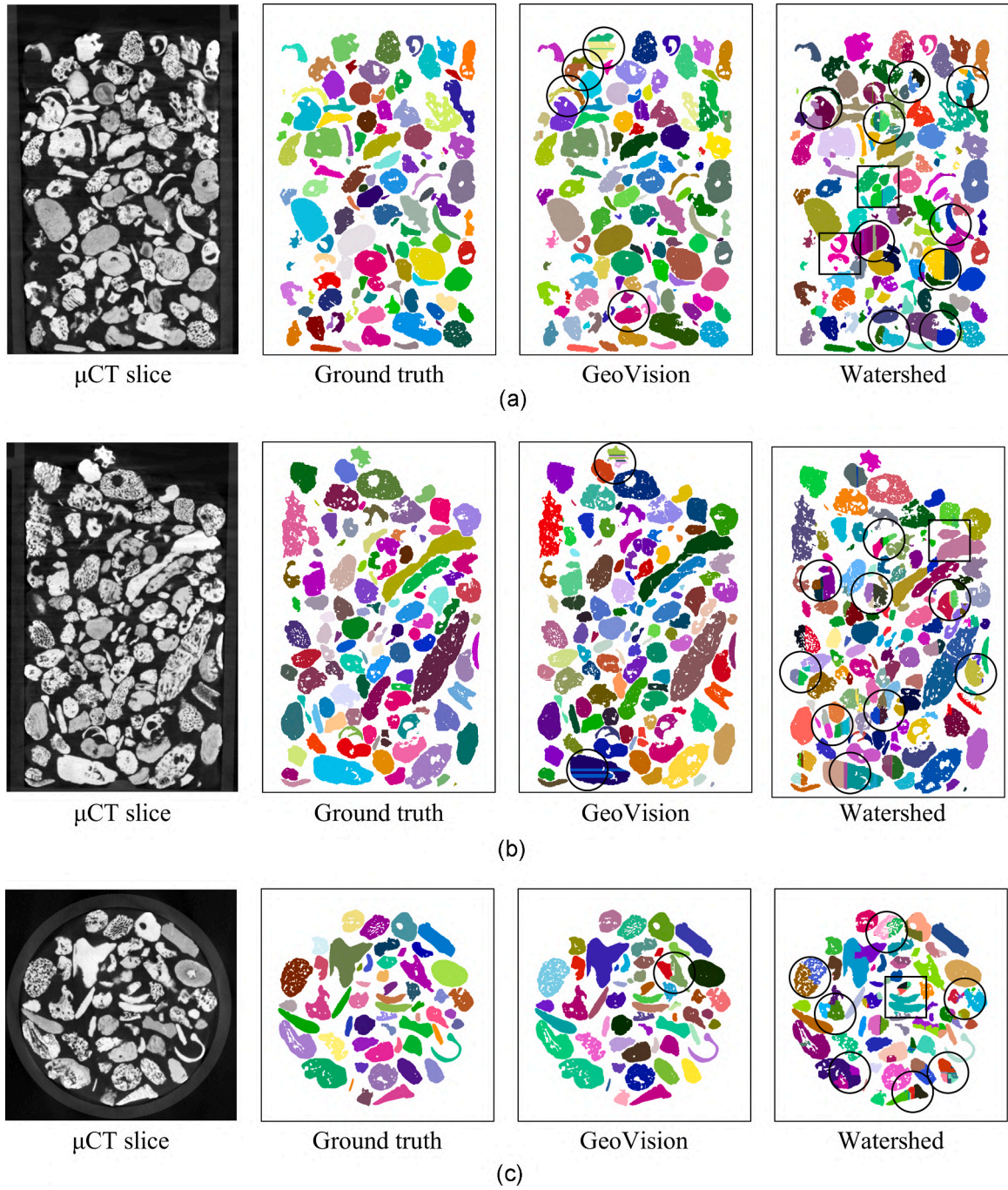


Fig. 17. 2D visualizations of representative slices along the x , y , and z -axis for CS2: (a) $x = 400$, (b) $y = 350$, and (c) $z = 528$.

inherently compatible with arbitrary grain shapes, owing to the strong mask extraction capabilities of the vision foundation model. GeoVision achieves a much better reconstruction, but at a disadvantage in computational time compared to the watershed. The total computational time for SAM-ViT-B, SAM-ViT-L, SAM-ViT-H, and watershed are 11,057 s, 13,471 s, 14,260 s, and 252 s, respectively. Although GeoVision has a longer computational time compared to the watershed method, it ensures satisfactory results and eliminates much of the tedious manual post-correction.

The 2D visualizations of representative slices parallel to the x , y , and z -axis for CS1 and CS2 are illustrated in Fig. 16 and Fig. 17, respectively. Note that the typical over-segmentations are marked by black solid circles while under-segmentation are denoted by black solid squares. These figures clearly demonstrate the superior performance of the

GeoVision framework compared to the watershed algorithm, as indicated by significantly fewer instances of over- and under-segmentation. Both segmentation errors are prevalent in the watershed method, especially over-segmentations; in contrast, GeoVision exhibits almost no under-segmentation due to robust mask identification of the vision foundation model, with only a minimal occurrence of over-segmentation. These issues of GeoVision are attributed to mask loss in one of the slices.

Fig. 18 provides an example for explaining the origin of 3D over-segmentation and the elimination of 2D over-segmentation. The failure to extract the mask in the black dashed square in slice $z = 194$ (Fig. 18b and c) results in the over-segmentation of the grain along the z -axis. This over-segmentation is evident in the slices along the x and y -axes, where a grain is split into two parts with a one-pixel-wide gap

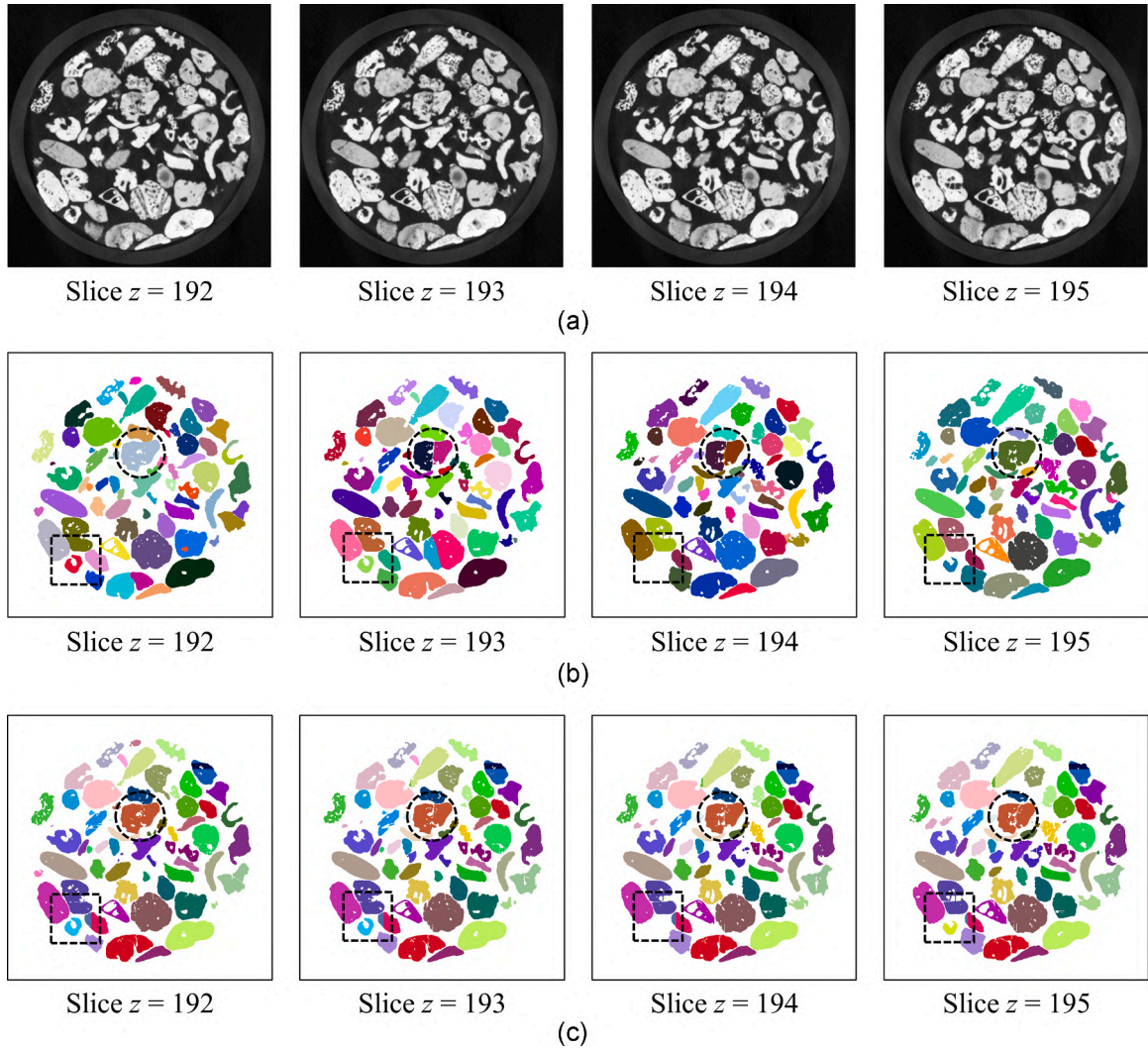


Fig. 18. Illustration for the origin of 3D over-segmentation and the elimination of 2D over-segmentation in GeoVision: (a) μ CT slices; (b) 2D mask map; and (c) 3D mask map.

between them, representing the missing mask. Consequently, the missing masks along the x and y -axes lead to over-segmentation in the slices along the z -axis. The underlying reason for this phenomenon is the inaccurate assignment of prompts, which prevents the vision foundation model from identifying the missing mask. A possible remedy could be the use of more advanced input prompts instead of grid points. For example, the iEDT can be modified to determine local minima and serve as the prompts to guarantee no masks are missed. Furthermore, the quality of μ CT images is also critical. If the image quality is poor (e.g., characterized by significant noise and low resolution), the VFM may struggle to accurately identify grains even with sufficient and reasonable prompts. Notably, the GeoVision framework can fix 2D over-segmentation errors. For instance, the 2D over-segmentation errors illustrated in the black dashed circle of Fig. 18b are corrected by the GrainStitch algorithm, as demonstrated in the corresponding 3D mask map (Fig. 18c).

4.5. Robustness and application

To demonstrate the robustness and application value of the proposed framework, the open-source dataset provided by Pinzón et al. [43] is utilized. This dataset consists of μ CT images obtained from five triaxial compression tests conducted at a confining pressure of 50 kPa on cylindrical specimens measuring 140 mm in height and 70 mm in

diameter, composed of slender green lentils. These specimens were scanned at regular intervals corresponding to 1 % axial strain (ϵ_a). For illustrative purposes, four scans of a specimen comprising 9408 lentils, with an initial orientation of 45 degrees (the max ϵ_a is 27 %), at ϵ_a values of 0 %, 10 %, 20 %, and 27 % are selected, as depicted in Fig. 19a. Despite the simplicity of lentil morphology, the substantial number of lentils in a single specimen serves as an effective benchmark for evaluating the robustness and efficiency of our framework. The spatial resolution of these μ CT images is $180 \times 180 \times 180 \mu\text{m}$. The dimensions of the images along the x , y , and z -axes for $\epsilon_a = 0 \%$, 10 %, 20 %, and 27 % are as follows: $400 \times 400 \times 780$, $420 \times 420 \times 700$, $500 \times 500 \times 620$, and $520 \times 520 \times 560$ voxels, respectively. The reconstructed results are obtained by model SAM-ViT-B at $n = 50$. The reference time for the sample at $\epsilon_a = 0 \%$, 10 %, 20 %, and 27 % is 5.1, 4.2, 6.1, and 6.5 hours, respectively. The ground truths and reconstructed results are shown in Fig. 19b and c, respectively. These results indicate a high alignment between reconstructions and ground truths by visual inspection, demonstrating the robustness of our approach preliminarily.

The quantitative assessments of our method in reconstructing four specimens are illustrated in Fig. 20. For these reconstructions, all metrics are closely aligned at high values. Specifically, the precision, recall, and AP values are approximately 0.98, 0.95, and 0.95, respectively, when the IoU matching threshold is set at 0.5. Even under the stringent matching criterion of $\text{IoU} = 0.9$, the precision, recall, and AP values

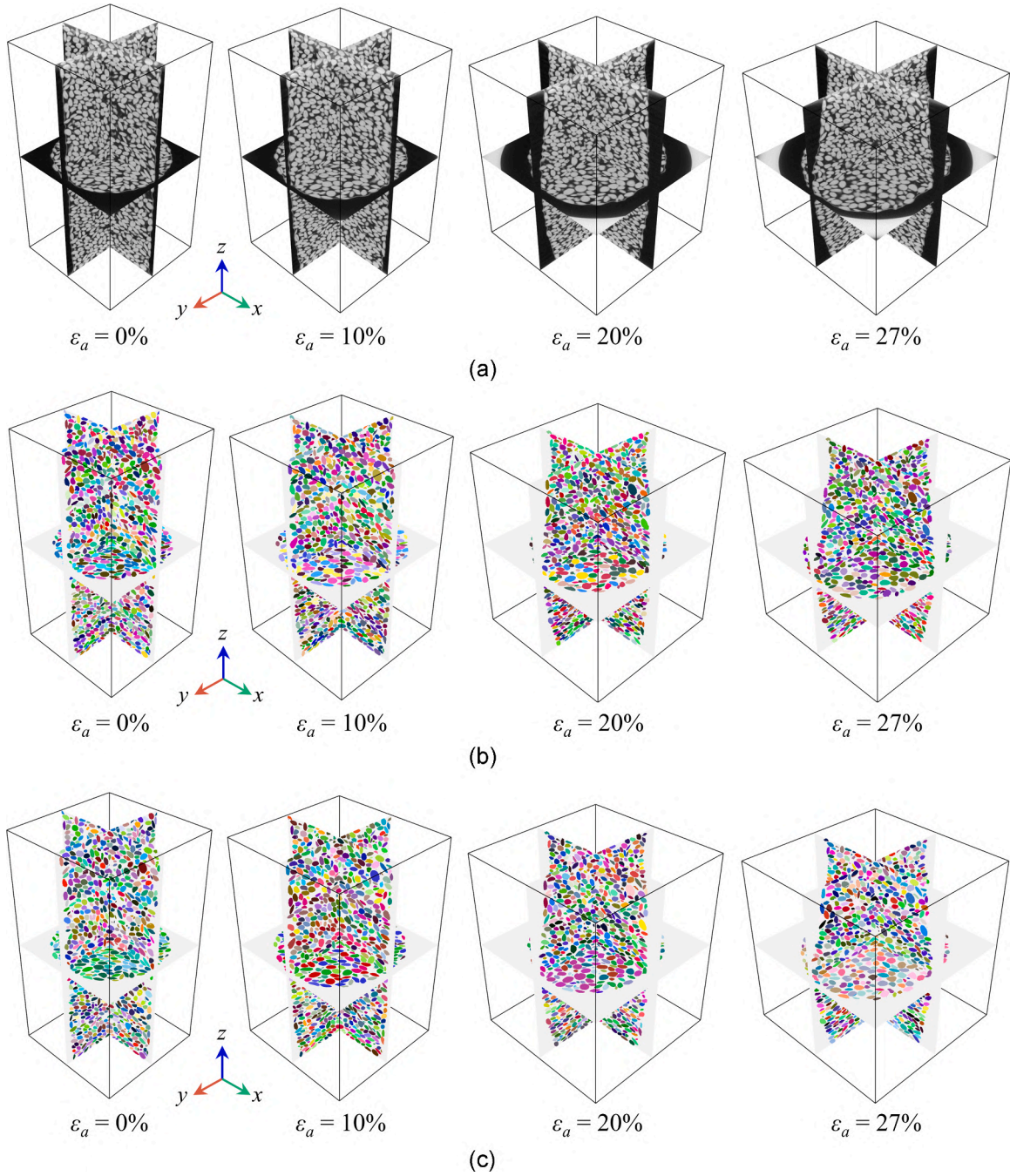


Fig. 19. Four lentil specimens: (a) μ CT images; (b) ground truth; and (c) reconstructions.

remain robust at 0.85, 0.84, and 0.75, respectively. These results underscore the robustness of the proposed framework in reconstructing 3D samples at varying states in experiments.

The purpose of 3D grain reconstruction is to probe microstructural characteristics such as interlocking, particle breakage, and anisotropy, thus optimizing geotechnical designs [44–46]. Therefore, evaluating reconstruction quality must incorporate microstructural properties rather than relying solely on accuracy metrics. In this study, the contact map which maps the coordination number Z (or number of contacts per grain) to grains by colors, is used to validate the application value of the proposed framework. The Python-based open-source software spam [47] is used for extracting contact maps based on reconstruction results. The mean coordination number $\bar{Z}$ within an assembly is defined as follows:

$$\bar{Z} = \frac{2N_c}{N_p} \quad (14)$$

where N_c and N_p denote the number of contacts and grains within the assembly, respectively. Fig. 21 displays the contact maps for both the ground truth and the reconstructed four specimens. The mean coordination number derived from our reconstruction is slightly higher than that of the ground truth, which may be attributed to over-segmentation errors. Interestingly, the mean coordination number experiences a decrease when $\varepsilon_a \leq 20\%$ and urgently increases until $\varepsilon_a = 30\%$. The decline in the early stage ($\varepsilon_a \leq 20\%$) is reasonable as grains are rearranged and packed, consistent with previous observations [48]. However, the increase in mean coordination number when $\varepsilon_a > 20\%$ is uncommon which may relate to the unique mechanical properties of

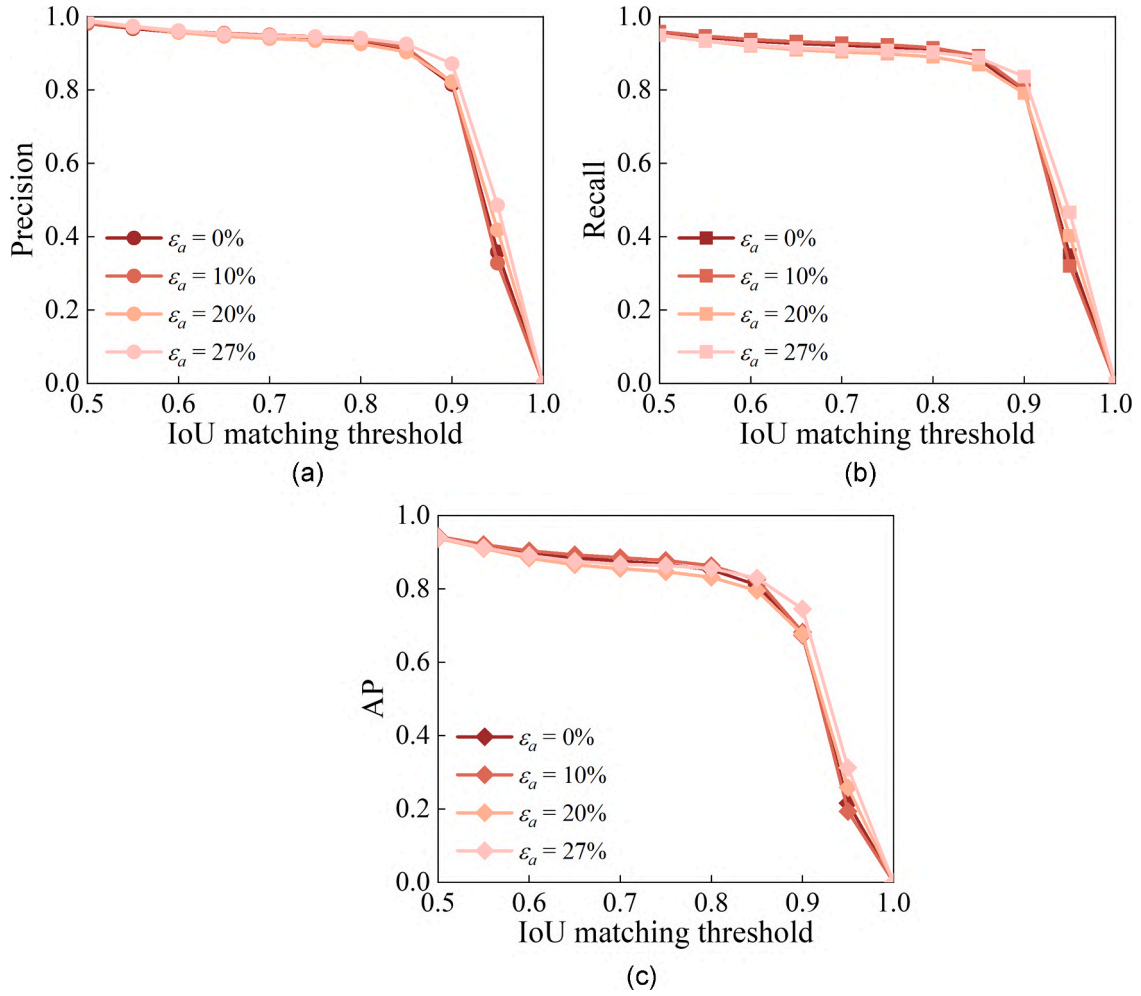


Fig. 20. Evaluation metrics at different IoU matching thresholds (from 0.5 to 1 in increments of 0.05) for reconstructing lentil specimens at different strains: (a) precision; (b) recall; and (c) AP.

lentils that needed further investigation but were not explored in this study. This result underscores the importance of 3D grain reconstruction in revealing complicated mechanisms of granular media. Notably, the distribution of coordination numbers for both results at the same strains is highly similar. These findings demonstrate the robustness of our proposed framework and its potential for application in tests aimed at exploring micromechanical properties.

5. Conclusions

This paper proposes a novel framework incorporating vision foundation models (VFMs) for the three-dimensional (3D) reconstruction of granular media from X-ray micro-computed tomography (μ CT) images. This framework was applied to reconstruct two carbonate sand samples with irregular morphologies and four lentil samples from an open-source dataset to verify its feasibility and robustness. VFMs, combined with a two-step strategy, enabled the extraction of high-quality two-dimensional (2D) masks denoting grains from μ CT images. 2D masks along x , y , and z -axes were transformed and mapped into one 3D mask map, through an optimal transport (OT)-based algorithm. Grains were then reconstructed using the marching cubes method and smoothed with Gaussian filtering.

The 3D reconstructions enhanced by VFMs demonstrated excellent agreement with the ground truth, effectively capturing interior textures and grain morphologies. VFMs possess a robust capability to identify arbitrary grain shapes from images, establishing our method as a generic

framework. The proposed framework outperformed the state-of-the-art method in reconstructing carbonate sands, achieving a 50 % improvement in accuracy and attaining 98 % average precision in reconstructing lentil samples. Under- and over-segmentation errors were notably less frequent in our reconstructions compared to the benchmark. The microstructural characteristics of lentil samples are extracted from our reconstructions and align with the ground truth well. Additionally, reconstruction accuracy increases with the prompt size until prompts are sufficient and influenced less by VFM size. The inference time linearly increases with prompt size. A grid size of 32×32 regular points, along with the most lightweight VFM, is recommended for initial trials to guarantee satisfactory results.

The primary limitation of the proposed framework is the relatively long reference time. To address this issue, future research will focus on establishing lightweight VFMs and optimizing prompts. Furthermore, various other types of granular media are also will be reconstructed to further demonstrate the robustness and application value of the proposed framework.

CRedit authorship contribution statement

Ruidong LI: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Zhen-Yu YIN:** Writing – review & editing, Supervision, Funding acquisition, Data curation. **Shao-Heng HE:** Writing – review & editing.

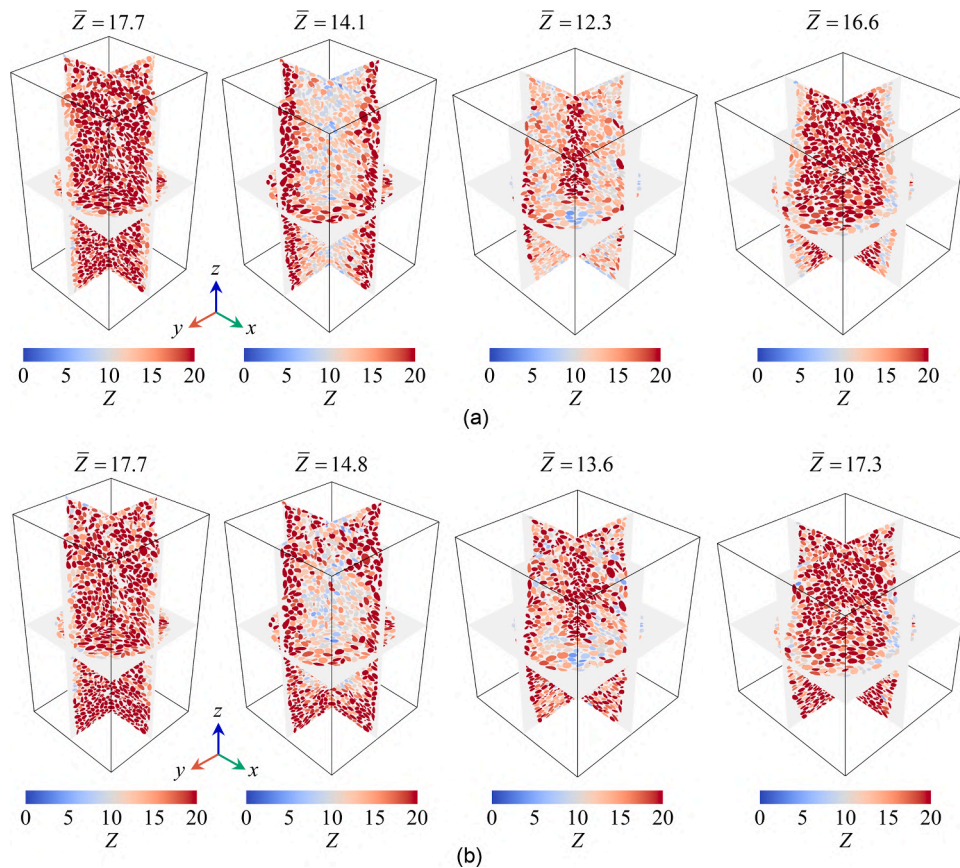


Fig. 21. Contact maps: (a) ground truth; and (b) reconstruction; Note from the left to right are specimens at $\varepsilon_a = 0\%$, 10% , 20% , and 27% .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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